

# The Moral Crumple Zone: Structural Conditions for Oversight Erosion in Human-AI Collaboration

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## Abstract

Madeleine Elish (2019)’s “moral crumple zone” describes a structural feature of complex human-automation systems in which the human operator absorbs responsibility for failures they lacked the practical capacity to prevent. This paper asks whether the antecedent conditions for that dynamic are observable before any particular failure occurs, and whether they are already present in everyday large language model use at scale. Using aggregate Claude.ai usage data covering one week in February 2026 we construct a crumple risk score combining AI autonomy, task failure rate, and human capability across 3,259 O\*NET task categories. We identify a small group of tasks, concentrated in professional work requiring contextual judgment, that exhibit this structural signature. The mechanism driving this concentration is not what three decades of automation bias research would suggest, with time savings bearing no meaningful relationship to oversight erosion. The operative variable is instead how the interaction is structured from the outset. Passive, directive engagement is a strong predictor of failure; active, learning-oriented dialogue predicts the opposite, and does so even among users who grant the AI more autonomy. The implication is that oversight quality is not a function of how much time users have to check AI output, but of whether the interaction format creates occasions for evaluative judgment at all.

## 1 Introduction

Someone must absorb the consequences when automation fails, but the distribution of this blame is rarely determined by who was actually in control. In 2019, Madeleine Elish introduced the “moral crumple zone” to describe a structural feature of complex human-automation systems: the human operator, regardless of their actual capacity to intervene, absorbs the moral and legal responsibility for system failures (Elish 2019). As a car’s crumple zone deforms on impact to protect the vehicle’s structural core, the human absorbs the force of failure to protect the system’s design from accountability. Elish presents a series of case studies (e.g., Chernobyl, Three Mile Island, Air France 447, Uber’s fatal autonomous vehicle crash) that each shared a crucial component in their architectures: nominal human oversight that was never intended to catch the failures it was blamed for missing. Each emerged in an environment where automated action became legible only after the fact. Accountability frameworks treated human presence as equivalent to human control, and regulation had not developed vocabulary for distributed agency.

This paper extends Elish’s framework to everyday human-AI interaction at scale, a domain where it has not yet been empirically examined, specifically with large language models in professional work contexts. We shift the analytical focus from blame attribution

after catastrophic failure to the upstream behavioral conditions that make failure more likely in the first place. The question thus becomes one of mitigation rather than one of blame.

We operationalize this using the Anthropic Economic Index (AEI), a publicly released dataset of aggregate usage statistics from Claude.ai covering one week in February 2026. The AEI classifies conversations by O\*NET occupational task, collaboration pattern, task success, AI autonomy level, among other characteristics, and reports statistics as aggregate cross-tabulations. Anthropic researchers created this structure<sup>1</sup> to protect users’ privacy, though it does impose some constraints on the nature of the analysis. While the data cannot support causal identification, it allows us to directly observe behavioral signals across thousands of distinct task types in a naturalistic setting.

## 2 Theoretical Framework

### 2.1 From Blame Attribution to Behavioral Preconditions

Elish’s original formulation is primarily concerned with accountability, and the crumple zone is a post-hoc phenomenon. After failure, the human operator is identified as the responsible party even when the system’s design renders their oversight nominal. This framing is powerful for legal and policy analysis but difficult to operationalize empirically. Blame attribution depends on institutional context—i.e., how regulators, courts, and other organizations construct accounts of failure after the fact—and is not directly observable in behavioral data before any particular failure occurs.

We adapt the concept by shifting the focus from blame to behavioral preconditions. If the crumple zone is a structural feature of human-automation systems, the conditions that produce it should be observable before failure. We consider a task to be in the crumple zone when three conditions are simultaneously present: (1) the AI is granted high autonomy; (2) the task has a high failure rate; and (3) a human could have completed the task independently. The third condition highlights where human capability is available but human oversight is not exercised, and is thus imperative in distinguishing the crumple zone from the limits of AI capabilities.

The AEI data cannot speak to how failures are accounted for after they occur nor how blame is attributed. It is, however, able to identify whether the three antecedent conditions co-occur for identifiable task categories and to characterize the mode of engagement associated with them. Paired with the collaboration pattern data, this allows claims about the structural preconditions at a scale post-hoc case analysis cannot match.

### 2.2 The Concept of Automation Bias and Its Limits

Automation bias literature, now spanning three decades and several thousand experiments, has documented with considerable consistency that humans over-rely on automated recommendations under cognitive load, time pressure, high task complexity, and increasing AI competence. The foundational taxonomy of omission and commission errors in human-automation systems originates with Skitka et al. (1999), with subsequent systematic reviews confirming these patterns across aviation, healthcare, and public administration (Goddard et al. 2012, Kücking et al. 2024). Horowitz and Kahn (2024)’s preregistered study across 9,000 adults in nine countries found that automation bias in national security task identification follows a non-linear curve with respect to AI knowledge, which they describe as a

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<sup>1</sup>See Tamkin et al. (2024) for more.

Dunning-Kruger effect for AI trust. Those with the least experience show slight algorithm aversion, bias increases at moderate knowledge levels, and levels off at the highest expertise. As moderate familiarity worsens the problem, this finding closes off the obvious solution of user education.

For LLMs specifically, intervention work has found that cognitive forcing functions reduce over-reliance more effectively than explanations or transparency displays, but that users consistently resist the interventions that help them most (Bućinca et al. 2021). Adding explanations to AI recommendations can paradoxically increase over-reliance by inflating perceived competence (Bansal et al. 2021). Uncertainty expressions reduce over-reliance on incorrect LLM outputs, but effects are highly phrasing-sensitive and do not generalize across tasks (Kim et al. 2024); broader intervention benchmarking finds that reductions in over-reliance in some conditions produce increases in under-reliance in others, with no universally effective design (Bo et al. 2025). These results all converge on the conclusion that informational and explanatory approaches to the calibration problem are structurally mismatched with the process they attempt to correct.

### 2.3 Trust Calibration and Its Structural Dimensions

Lee and See (2004)’s foundational review of trust in automation provides the conceptual framework for understanding why the failure of the time savings mechanism is both surprising and theoretically significant. The authors argue that three distinct cognitive processes—analytic, analogical, and affective—are responsible for calibrating users’ trust in automation, but that the affective process systematically dominates. Trust is shaped less by rational assessment of a system’s track record than by the perceived competence and interpretability of the system. Informational interventions targeting rational calibration, such as explainability interfaces or performance feedback, are therefore working against a process that has already run.

Carnat (2024) extends this argument to generative LLMs specifically, arguing that their conversational fluency creates conditions for something she calls “cognitive automation.” The system’s output feels like it was produced by a competent colleague rather than a statistical process, such that users do not activate independent judgment. The relevant feature of LLMs is that their failures are fluent; they produce confident, coherent, contextually plausible output that requires domain knowledge to identify as wrong. This is a structurally different failure mode from earlier automated systems, where malfunction is typically forced to be visible. In Elish’s cases, the automation produced observable states, such as instrument readings and system alerts, that a skilled person could in principle interpret correctly. The failures thus were ones of interface design and cognitive overload, in which those responsible were ill-positioned to synthesize and act on the available information in time. The failure mode in LLM interactions, however, is epistemically prior. Outputs appear inherently reliable, and there is no guarantee the user knows there is anything to check.

Tankelevitch et al. (2024) situate this within a broader argument that generative AI systems impose qualitatively new metacognitive demands at every interaction stage, from prompt formulation to output evaluation, that users are systematically underprepared to supply. The demand is both cognitive and structural in that the interaction format shapes what kinds of scrutiny are natural to apply.

## 2.4 Collaboration Mode as a Candidate Mechanism

The AEI’s collaboration pattern data offer a different analytical lever. Each conversation in the AEI is classified into one of five modes: *directive* (the user issues instructions and accepts output with minimal back-and-forth), *learning* (exploratory dialogue that builds understanding), *task iteration* (repeated refinement of AI output), *feedback loop* (evaluative input that shapes subsequent output), and *validation* (checking AI work against an external standard). These categories, defined by Anthropic’s internal classifier, map onto a natural spectrum from passive delegation to active engagement.

The premise of the crumple zone predicts that directive interactions should be associated with worse outcomes. In Elish’s terms, the user has nominally “overseen” the task by initiating it and accepting its output, but has not exercised the judgment that would catch errors expressed with unwarranted confidence. The directive share of tasks should correlate positively with failure rates, while the learning and iteration share should correlate negatively. Research has found that in single-transaction formats (AI displays output, human accepts or rejects), human-AI pairs frequently underperform the AI working alone (Gomez et al. 2024). Decision-oriented dialogue, in which iterative exchange surfaces constraints the single-transaction format abstracts, substantially outperforms single-turn interaction even when controlling for dialogue length (Lin et al. 2024). These results motivate the hypothesis that collaboration mode, not time savings, is the operative variable.

## 3 Data and Methods

### 3.1 The Anthropic Economic Index as an Analytical Object

The fifth release of the Anthropic Economic Index (Massenkoff et al. 2026) covers Claude.ai Free and Pro usage from February 5–12, 2026. Structured in long format, each row represents a single metric value for a specific geographic unit, analytical facet, and category combination. For the purposes of this analysis, we focus on global-level statistics. We extracted three analytical layers from the data: categorical facet percentages (collaboration patterns, task success, and related dimensions), numeric facet summaries (autonomy, time estimates, education levels), and joint distributions cross-tabulating two dimensions. For example, `onet_task::collaboration` provides the collaboration pattern breakdown for each O\*NET task, while `onet_task::task_success` provides the success rate for each task. These intersections are the primary analytical object, allowing each task to be characterized on multiple dimensions simultaneously without requiring individual-level data. The dataset contains 3,259 O\*NET tasks with autonomy data, of which 1,392 have sufficient observations during the covered period. *Sufficient observations* is defined as having a raw count of autonomy observations per task greater than the overall median of 65. At the request level, 730 bottom-up request categories are available, providing more semantically coherent groupings than the occupational definitions.

### 3.2 Crumple Risk Scoring

A score of *human ability* is provided for each O\*NET task, estimating the mean ability of a human to independently complete the task successfully. The core analytical challenge is combining AI autonomy, failure rate, and human ability into a single risk score that is robust to the noise inherent in aggregate statistics with variable sample sizes. We address

this through empirical Bayes shrinkage followed by rank-percentile normalization. For each of the three inputs, a shrinkage factor is derived entirely from the data:

$$\text{shrinkage} = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2} \quad (1)$$

For autonomy, the within-task variance is recovered from the 95% confidence interval as  $(CI_{\text{width}}/3.92)^2$ , and the between-task variance is estimated by method-of-moments as the variance of observed means minus the mean within-task variance. For the binomial variables (failure rate and human ability), within-task variance is  $p(1-p)/n$  using the observed proportion and task observation count. The shrunk estimate for each task is:

$$\hat{\theta}_{\text{shrunk}} = \bar{\theta} + \text{shrinkage} \times (\theta_{\text{obs}} - \bar{\theta}) \quad (2)$$

This pulls extreme estimates from low-sample tasks toward the global mean in proportion to their uncertainty. Tasks observed across thousands of conversations retain their observed values nearly unchanged, while tasks observed in a handful are regularized substantially. The shrinkage diagnostics confirm this is working as intended, with autonomy showing  $r(\text{obs}, \text{shrunk}) = 0.983$  with 83% range compression, failure showing  $r = 0.9906$  with 96% range compression, and human ability showing  $r = 0.875$  with 61% range compression.

After shrinkage, all three variables are converted to rank percentiles, producing values uniformly distributed on  $[0, 1]$ . The crumple risk score is then:

$$\text{crumple risk} = \text{autonomy rank} \times \text{failure rank} \times \text{human ability rank} \quad (3)$$

Multiplication produces near-zero scores whenever any single component is near zero, which is precisely the theoretical requirement, whereas addition would allow high scores even when one condition is absent. Both a task with extreme autonomy but zero failure and a task with high failure but minimal autonomy will score near zero.

### 3.3 Quadrant Analysis

To complement the continuous score, tasks are partitioned into four quadrants by median splits on AI autonomy (3.43) and failure rate (27.9%):

Table 1: Task quadrants by AI autonomy and failure rate.

|               | Low failure                       | High failure                      |
|---------------|-----------------------------------|-----------------------------------|
| High autonomy | AI-competent ( $n = 396$ , 28.4%) | Crumple risk ( $n = 300$ , 21.6%) |
| Low autonomy  | Routine ( $n = 300$ , 21.6%)      | Struggling ( $n = 396$ , 28.4%)   |

Autonomy and failure are negatively correlated overall ( $r = -0.178$ ), meaning Claude is more successful when granted more autonomy. The crumple risk tasks are those that violate this general trend, cases in which users grant more autonomy than the model’s competence warrants.

### 3.4 Hypotheses

The theoretical framework motivates three testable hypotheses:

- H1** Tasks exhibiting the simultaneous structural signature of high AI autonomy, high failure rate, and high human capability will be concentrated in professional work requiring contextual judgment, rather than distributed uniformly across task types.
- H2** Time savings will correlate positively with AI autonomy and failure rate, such that tasks where AI saves the most time show the greatest oversight erosion.
- H3** Directive engagement will correlate positively with failure rates, while learning-oriented and iterative engagement will correlate negatively, at both the task and request level.

## 4 Results

### 4.1 H1: Task Risk Distribution

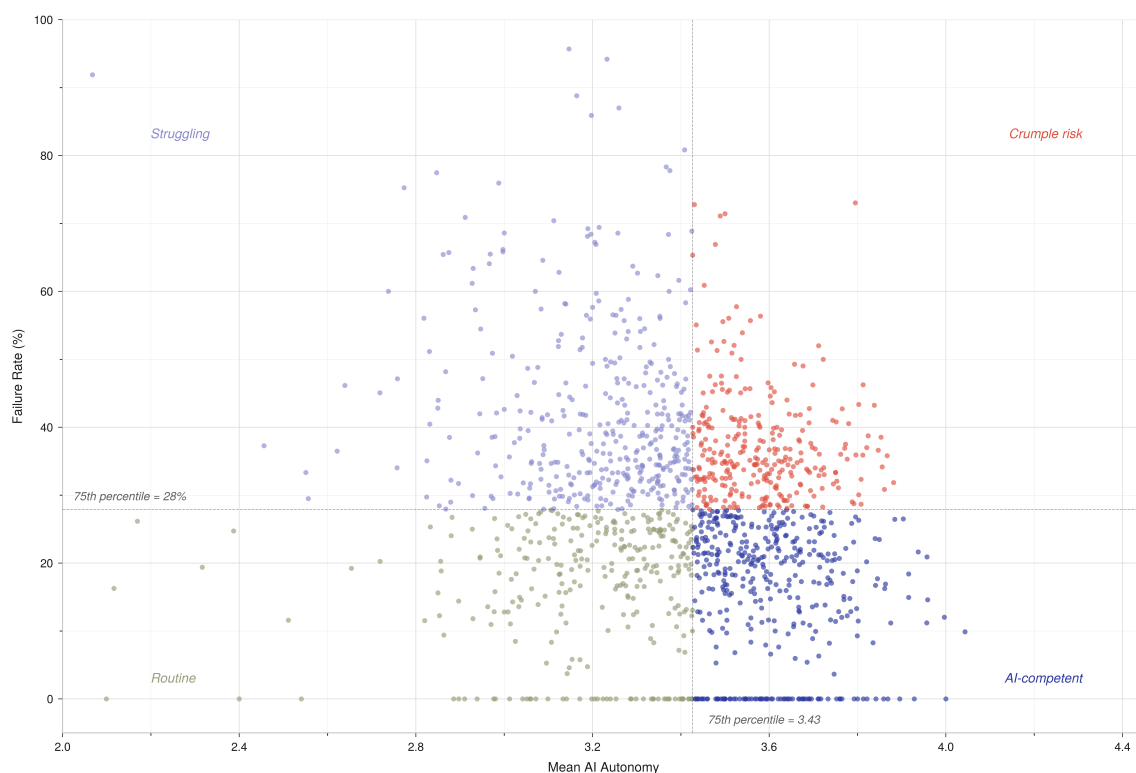


Figure 1: Quadrant scatter of AI autonomy versus failure rate for all O\*NET tasks. Crumple risk tasks (upper right) combine high autonomy with high failure.

Among the tasks with the highest risk of “crumple” were: *“assess project feasibility by analyzing technology, resource needs, and market demand,” “recommend, select, and help locate or obtain merchandise based on customer needs and desires,” “develop or implement marketing strategies,”* and *“give advice on sightseeing and shopping.”* These each notably require synthesizing incomplete information against a background of expertise and exercising judgment about what a specific situation calls for.

While one might expect the highest-risk tasks to be the most technically demanding, empirically they are the ones most dependent on professional judgment and expertise. AI

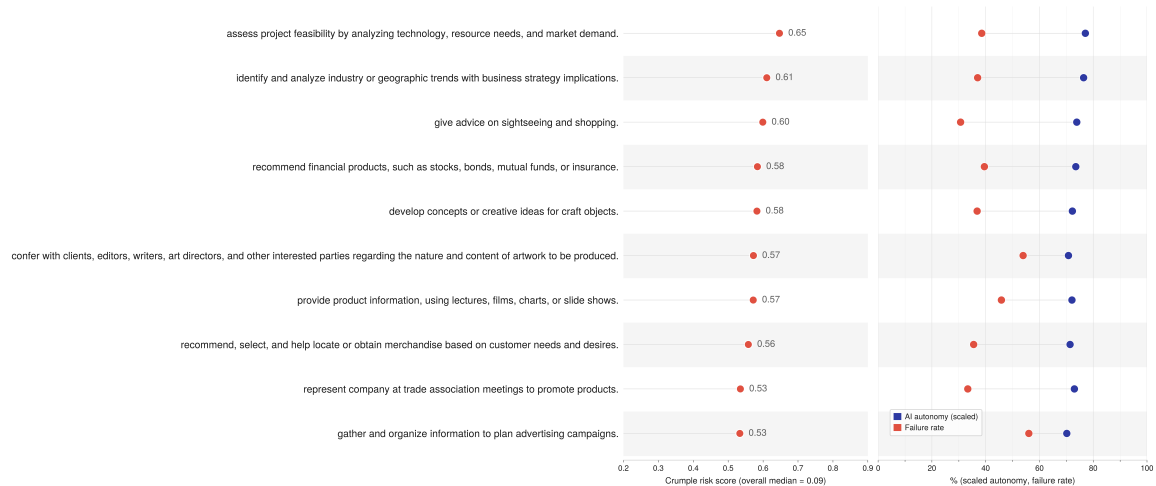


Figure 2: Highest-scoring O\*NET tasks on the crumple risk index.

generates a plausible response to an abstracted version of the task, but the failure takes place somewhere in the gap between that abstraction and the situation a competent professional would have treated as a constraint from the start.

In contrast, the bottom of the distribution is instructive. Among the lowest-scoring tasks are data entry, transcription, record maintenance, and software modification. These are tasks where the AI’s output is either immediately visible or detectable without domain expertise and where human ability is often constrained by design. They are also, notably, tasks where directive engagement is appropriate.

## 4.2 H2: Time Savings

The time savings hypothesis was not strongly supported. The correlation between time savings ratio and AI autonomy is +0.017; between time savings and failure rate, +0.048; between time savings and crumple risk, +0.074. Although this is not the result we expected, it is nonetheless an important result due to its implications for design strategy. Time pressure is among the most robust amplifiers of automation bias in the experimental literature, and the intuition that faster completion leads to less careful review is embedded in how oversight is typically modeled. If the task is already done and oversight is costly, the marginal return to checking falls. The AEI data suggest that this mechanism, compelling in controlled experiments or more “classical” forms of automation, may not be as powerful in consumer applications of LLMs.

Several potential explanations may clarify the dissonance. Task difficulty may be a common driver of both time savings and oversight quality in ways the aggregate data cannot disentangle. Tasks that take longer to complete and offer less time savings may also be tasks at which AI is more likely to fail or in which careful review is less likely for independent reasons. A second interpretation is that time pressure may be able to detect automation bias within a particular task type, but unable to predict which task types are most prone to automation bias. A third interpretation, supported by the collaboration data, is that oversight quality is determined not by the time available to check but by the engagement mode that structures the interaction from the outset. Users who engage directly have already, at the outset, structured the interaction in a way that makes careful review unlikely,

and time savings is therefore downstream of this structural choice.

### 4.3 H3: Collaboration Patterns

The collaboration pattern data tells a different story. At the request level, the correlation between directive interaction share and failure rate is  $r = +0.493$ . The correlation between learning mode share and failure rate is  $r = -0.336$  and task iteration is similarly protective ( $r = -0.282$ ).

Table 2: Correlation of collaboration patterns with task outcomes.

| Pattern        | vs. failure | vs. autonomy |
|----------------|-------------|--------------|
| Directive      | +0.493      | -0.489       |
| Feedback loop  | +0.298      | +0.195       |
| Learning       | -0.336      | +0.293       |
| Task iteration | -0.282      | -0.020       |
| Validation     | -0.146      | -0.160       |

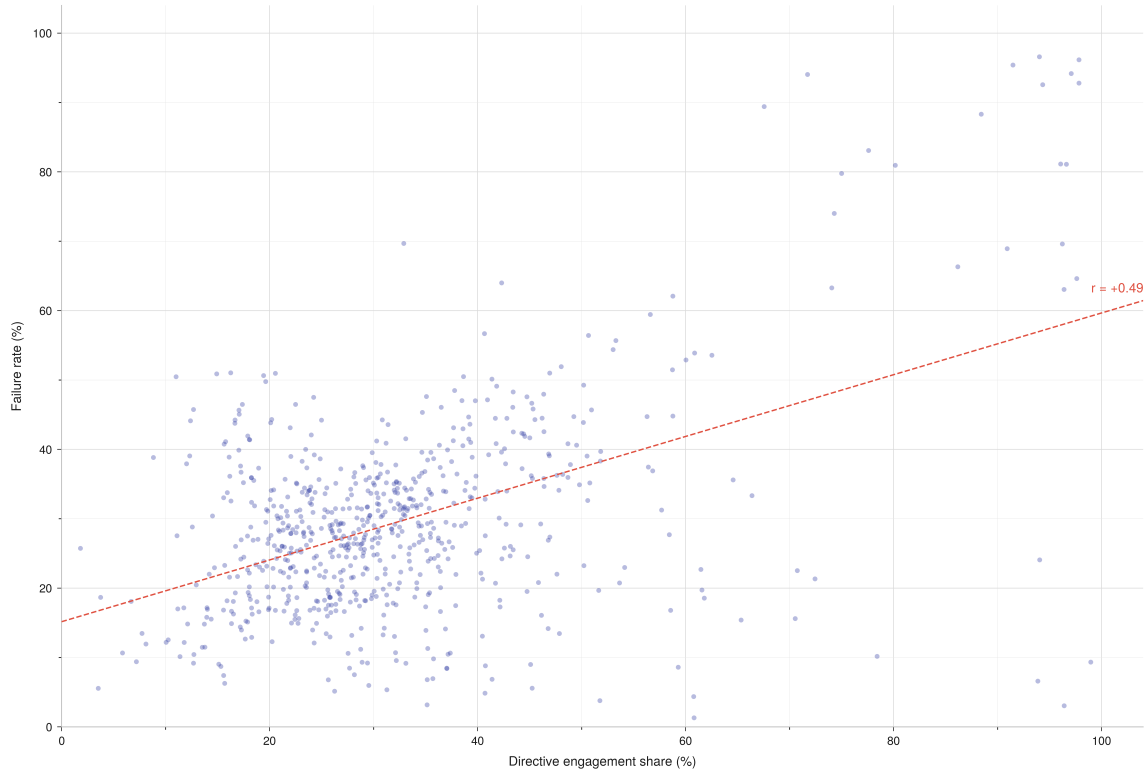


Figure 3: Scatterplot of directive interaction share versus task failure rate across request categories.

The pattern at the quadrant level reinforces this, with crumple risk tasks exhibiting 37% directive share and 22% learning share. For comparison, AI-competent tasks, which have only slightly higher autonomy levels, show 26% directive and 44% learning. The two quadrants differ primarily in how they structure interactions around similar levels of autonomy.



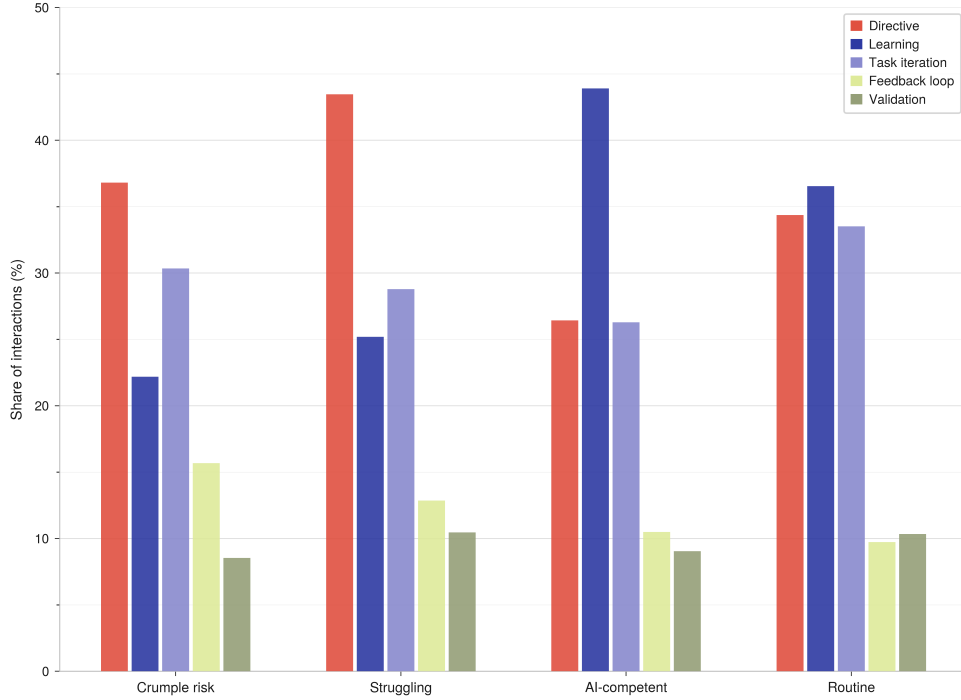


Figure 4: Collaboration pattern breakdown by task quadrant.

The quartile comparison is perhaps the most direct statement of the finding. Request categories in the top quartile of directive share have a mean failure rate of 38.6% and mean autonomy of 3.28. Request categories in the top quartile of learning share have a mean failure rate of 22.1% and mean autonomy of 3.48. More delegation with more engagement produces better outcomes than less delegation with less engagement.

## 5 Discussion

### 5.1 The Collaboration Paradox

Earlier automated systems like the ones mentioned by Elish produced observable states, such as system alerts, that a skilled person could in principle interpret. The failure was one of interface design and cognitive overload, a question of whether those responsible could synthesize available information in time. LLM failures are epistemically prior to that. There is no instrument reading to misread because the system produces fluent, plausible text that requires domain expertise to identify as wrong. Tasks enter the “crumple zone” because they are where the gap between what the AI models and what the situation requires is both largest and least visible to a user who has not engaged evaluatively.

While it seems paradoxical for directive collaboration to have lower autonomy and higher failure rates than learning-style collaboration, it is not surprising given earlier research; AI autonomy is known to be associated with better outputs, as are iterative interactions (Gomez et al. 2024, Lin et al. 2024). However, the association of the two means more delegation with more engagement produces better outcomes than less delegation with less engagement. The explanation, we argue, is structural rather than motivational. Directive engagement follows the structure of tool use wherein the user specifies an input and accepts an output.

There is no natural moment at which to insert evaluative judgment because the interaction is structured as a single transaction. Learning mode, by contrast, follows the structure of dialogue. Each prompt and response exchange generates new information that can surface the AI’s assumptions, flag unexpected framings, and prompt the user to apply contextual knowledge that was not in the original specification.

Current LLM interfaces make directive engagement the path of least resistance as the conversational format is architecturally a directive interface. The affordances that would make learning-mode engagement natural (explicit hypothesis refinement, comparative output review) are available in principle but require deliberate effort to use. Tankelevitch et al. (2024) argue that generative AI demands a stack of simultaneous metacognitive processes, among these task decomposition, reasonable confidence in prompting and evaluation abilities, and flexibility in prompting approach to receive desired output. Where those resources are not available, directive engagement may be less a preference than a ceiling, which shifts the design problem from nudging users toward learning-mode to building interfaces that distribute the metacognitive load rather than requiring users to supply it independently.

In a similar vein, proposals to improve AI oversight by increasing transparency, displaying confidence scores, or providing explanations target the rational-analytic channel of trust calibration that Lee and See (2004) demonstrate is not the dominant one. They assume that users who have correct information about AI reliability will adjust their oversight accordingly. The collaboration data suggest instead that what determines oversight quality is how the interaction is structured around that reliability. Design interventions that make learning-mode engagement the natural path through complex, judgment-intensive tasks without degrading the efficiency that makes LLMs valuable for routine tasks would be targeted at the mechanism the data actually implicate.

## 5.2 Why the Time Savings Null Reorients the Design Problem

A persistent strand of AI oversight thinking has organized itself around pace. If AI moves faster than human cognition, friction (mandatory review pauses, deliberate slowdowns) becomes the natural corrective. The time savings null forecloses this framing directly. Tasks where AI saves the most time are no more likely to show oversight erosion than any other. The distribution of crumple risk is orthogonal to the time savings distribution. If the mechanism were speed, we would expect a gradient across that distribution. Friction-based interventions, even if they worked as intended, would be imposing costs on the wrong variable.

The findings are also in line with Anthropic’s December 2025 study of 132 of its own engineers and researchers (Huang et al. 2025), which found that reported time savings were highly variable and, for a substantial minority, negative. Engineers who spent more time on Claude-assisted tasks explained that they had to debug AI-generated code or shoulder cognitive overhead for understanding output they didn’t write themselves. Claude’s main effect on productivity seemed to be on output volume, with engineers producing substantially more across task categories even when per-task time held steady or increased. Crucially, the engineers who reported the clearest time savings were those who had developed what the study calls “delegation intuitions”—strong informal rules for which tasks were quickly verifiable and which required active engagement. The time savings, where they existed, were downstream of strategic task selection rather than upstream of oversight quality.

### 5.3 Design Challenges and Limitations

These findings sit in productive tension with what users say they want. A qualitative study of 81,000 Claude users conducted by Anthropic in December 2025 (Huang et al. 2026) found that the largest group (19%) sought “professional excellence”—wanting AI to handle mundane tasks so they could focus on higher-level work—and another 14% sought cognitive offloading of administrative burden. The dominant user aspiration is precisely directive engagement, but directive delegation without evaluative engagement is structurally hospitable to crumple risk. A separate 17% of interviewees described their most valuable AI experience as cognitive partnership, and the request categories that show the best outcomes under high autonomy are exactly those where learning and iteration dominate. The design challenge is to make cognitive partnership the natural path through tasks that require it, without degrading the efficiency of directive engagement for tasks where it is appropriate.

Several considerations suggest this problem is unlikely to self-correct. Horowitz and Kahn’s (2024) finding that moderate AI familiarity increases rather than decreases automation bias means the problem is not likely to diminish as users gain experience. Carnat’s (2024) argument that LLM fluency creates specifically potent conditions for cognitive automation suggests it may intensify as models improve, as higher-quality outputs may be increasingly difficult to scrutinize. The trust calibration literature’s consistent finding that informational interventions do not reliably reduce over-reliance (Lee and See 2004) precludes that as an alternative. For judgment-intensive, high-autonomy tasks, the most natural path through the interaction should invite evaluative engagement through affordances that make learning-mode interaction as available as the single-transaction format that currently dominates.

As discussed above, the nature of the data inherently introduces limitations to these findings. Cross-sectional aggregates prohibit causal identification, user-level trajectory tracing, and control for unobserved confounders. The collaboration-outcome correlation does not establish that engagement mode causes outcome differences. Tasks that are inherently more tractable may naturally invite learning-mode engagement, and users with more relevant expertise may both engage in learning mode and achieve better outcomes for independent reasons. The aggregate AEI data can identify structural patterns; adjudicating between structural and selection accounts of those patterns requires more information not publicly available. The crumple risk score’s multiplicative structure is a theoretical assumption, as interactions between groups of three variables are not observed. The collaboration pattern labels are Anthropic’s internal classifications, and their validity as measures of the theoretical constructs they are intended to capture is assumed rather than established.

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